The documentation and process conversion measures necessary to comply with this revision shall be completed by 30 July 1993.

INCH-POUND

MIL-S-19500/514A
31 March 1993
SUPERSEDING
MIL-S-19500/514(EL)
25 March 1976

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER
TYPES 2N6274 AND 2N6277, JANTX AND JANTXV

This specification is approved for use by the U.S. Army Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN silicon, power transistors. Two levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings. T_C = +25°C, unless otherwise specified.

Type	P_T 1/ T_C = +25°C	P_T 1/ T_C = +100°C	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_J and T_{STG}	$R_{\theta JC}$
	W	W	V dc	V dc	V dc	A dc	A dc	°C	°C/W
2N6274	250	143	120	100	6	20	50	-65 to +200	.70
2N6277	250	143	180	150	6	20	50	-65 to +200	.70

1/ Derate linearly 1.43 W/°C between T_C = +25°C and T_C = +200°C.

1.4 Primary electrical characteristics. T_C = +25°C.

	h_{FE3} 1/	h_{FE2} 1/	$V_{BE(SAT)}$ 1/	$V_{CE(SAT)}$ 1/	C_{obo}	$\|h_{fe}\|$	Switching	
	V_{CE} = 4 V dc I_C = 50 A dc	V_{CE} = 4 V dc I_C = 20 A dc	I_C = 20 A dc I_B = 2 A dc	I_C = 20 A dc I_B = 2 A dc	V_{CB} = 10 V dc I_E = 0 f = 1 MHz	V_{CE} = 10 V dc I_C = 1 mA dc f = 10 MHz	t_{on}	t_{off}
			V dc	V dc	pF		µs	µs
Min	10	30				3		
Max		120	1.8	1.0	600	12	0.5	1.05

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Research Laboratory ATTN: AMSRL-EP-RD Fort Monmouth, NJ 07703-5601, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and on figure 1 herein.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750 and MIL-S-19500. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

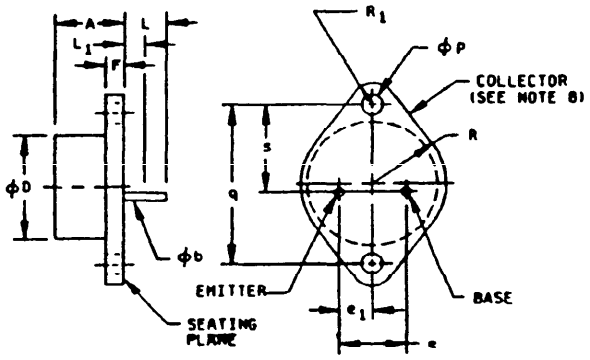
4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.250	.328	6.35	8.33	
ϕb	.057	.063	1.45	1.60	5, 9
ϕD	---	.875	---	22.23	3
e	.420	.440	10.67	11.18	4
e_1	.205	.225	5.21	5.72	4, 5
F	.060	.135	1.52	3.43	
L	.312	.500	7.92	12.70	5
L_1	---	.050	---	1.27	5, 9
ϕ_p	.151	.161	3.84	4.09	7
q	1.177	1.197	29.90	30.40	
R	.495	.525	12.57	13.34	
R_1	.131	.188	3.33	4.78	6
s	.655	.675	16.64	17.15	4



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Body contour is optional within zone defined by ϕD.
4. These dimensions shall be measured at points .050 inch (1.27 mm) to .055 inch (1.40 mm) below seating plane. When gauge is not used, measurement shall be made at seating plane.
5. Both terminals.
6. At both ends.
7. Two holes.
8. The collector shall be electrically connected to the case.
9. ϕb applies between L_1 and L. Diameter is uncontrolled in L_1.

FIGURE 1. Physical dimensions.

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4.3 Screening (JANTX and JANTXV levels). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement JANTX and JANTXV levels only
1/	Thermal response (see 4.3.2)
9	Not applicable
11	h_{FE2}; I_{CEX1}
12	See 4.3.1
13	ΔI_{CEX1} = 100 percent of initial value or 1 μA dc; whichever is greater; Δh_{FE2} = ±25 percent of initial value; subgroup 2 of table I herein.

1/ This test shall be performed anytime between screens 3 and 9.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_J = 187.5°C ±12.5°C; V_{CB} ≥ 20 V dc.

4.3.2 Thermal response (ΔV_{BE} measurements). The ΔV_{BE} measurements shall be performed in accordance with MIL-STD-750, method 3131. The ΔV_{BE} conditions (I_H and V_H) and maximum limit shall be derived by each vendor. The chosen V_{BE} measurement and conditions for each device in the qualification lot shall be submitted in the qualification report and a thermal response curve shall be plotted. The chosen ΔV_{BE} shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. One hundred percent Safe Operating Area (SOA) testing may be performed in lieu of thermal response testing herein provided that the appropriate conditions of temperature, time, current, and voltage to achieve die attach integrity are submitted to the qualifying activity. The following parameter measurements shall apply:

a. I_H measurement . . . . . 10 mA.

b. V_{CE} measurement voltage . . . . . 20 V (same as V_H).

c. I_H collector heating current . . . . . 1 A (minimum).

d. V_H collector-emitter heating voltage . . . . . 20 V (minimum).

e. t_H heating time . . . . . 100 ms.

f. t_{MD} measurement delay time . . . . . 50 to 80 μs.

g. t_{SW} sample window time . . . . . 10 μs (maximum).

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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table IVb of MIL-S-19500. Subgroup 5: Thermal resistance; method 3131; conditions, see 4.5.2.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

a. Subgroup 2: Terminal strength (lead fatigue), method 2036, conditions: Test condition A, weight = 10 pounds, t = 15 s.

b. Subgroup 3: Shock, method 2016, conditions: Nonoperating, 1,500 g's, 0.5 ms, 5 blows in each orientation X1, Y1, Y2, and Z1. Constant acceleration; method 2006; conditions, 5,000 g's, in each orientation X1, Y1, Y2, and Z1.

4.5 Method of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power applications shall be 2 A dc, minimum.

b. Collector to emitter voltage magnitude ≥ 16 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be 25°C ≤ T_R ≤ 75°C and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to case.

f. Maximum limit shall be $R_{\theta JC}$ = 0.7°C/W.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Collector to emitter breakdown voltage	3011	Bias condition D, I_C = 50 mA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N6274				100		V dc
2N6277				150		V dc
Collector to emitter cutoff current	3041	Bias condition D	I_{CEO}		50	µA dc
2N6274		V_{CE} = 50 V dc				
2N6277		V_{CE} = 75 V dc				
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = -1.5 V dc,	I_{CEX1}		10	µA dc
2N6274		V_{CE} = 120 V dc,				
2N6277		V_{CE} = 180 V dc				
Emitter to base voltage cutoff current	3061	Bias condition D, V_{EB} = 6 V dc	I_{EBO}		100	µA dc
Collector to base cutoff current	3036	Bias condition D	I_{CBO}		10	µA dc
2N6274		V_{CB} = 120 V dc				
2N6277		V_{CB} = 180 V dc				
Base emitter saturated voltage	3066	Test condition A, I_C = 20 A dc, I_B = 2.0 A dc, pulsed (see 4.5.1)	$V_{BE(sat)}$		1.8	V dc
Collector-emitter saturated voltage	3071	I_C = 20 A dc, I_B = 2.0 A dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		1.0	V dc
Collector-emitter saturated voltage	3071	I_C = 50 A dc, I_B = 10 A dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		3.0	V dc
Forward-current transfer ratio	3076	V_{CE} = 4 V dc, I_C = 1 A dc, pulsed (see 4.5.1)	h_{FE1}	50		
Forward-current transfer ratio	3076	V_{CE} = 4 V dc, I_C = 20 A dc, pulsed (see 4.5.1)	h_{FE2}	30	120	
Forward-current transfer ratio	3076	V_{CE} = 4 V dc, I_C = 50 A dc, pulsed (see 4.5.1)	h_{FE3}	10		

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current 2N6274 2N6277	3041	Bias condition A, V_{BE} = -1.5 V dc, V_{CE} = 120 V dc, V_{CE} = 180 V dc	I_{CEX2}		1.0	mA dc
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 4.0 V dc, I_C = 20 A dc, pulsed (see 4.5.1)	h_{FE4}	10		
Subgroup 4						
Pulse response	3251	Test condition A, except test circuit and pulse requirements See figure 2 herein.				
Turn-on time		V_{CC} = 80 V dc, I_C = 20 A dc, I_{B1} = 2 A dc	t_{on}		0.5	µs
Turn-off time		V_{CC} = 80 V dc, I_C = 20 A dc, I_{B1} = $-I_{B2}$ = 2 A dc	t_{off}		1.05	µs
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc, I_C = 1 A dc, f = 10 MHz	$\|h_{fe}\|$	3	12	
Open circuit output capacitance	3236	V_{CB} = 10 V dc, I_E = 0, f = 1.0 MHz	C_{obo}		600	pF

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +25°C, t = 1 s, 1 cycle, (see figure 3)				
Test 1		V_{CE} = 5 V dc, I_C = 50 A dc				
Test 2		V_{CE} = 8.6 V dc, I_C = 29 A dc				
Test 3						
2N6274		V_{CE} = 80 V dc, I_C = 165 mA dc				
Test 4						
2N6277		V_{CE} = 120 V dc, I_C = 110 mA dc				
Safe operating area (switching)	3053	Load condition C (unclamped inductive load) (see figure 4), T_C = +25°C, duty cycle ≤ 10 percent, R_S = 0.1Ω, $t_r = t_f \leq 500$ ns				
Test 1		t_p approximately 5 ms (vary to obtain I_C), R_{BB1} = 2Ω, V_{BB1} = 12 V dc, R_{BB2} = ∞, V_{BB2} = 0, V_{CC} = 50 V dc, I_C = 40 A dc, L = 100 µH 2/				
Test 2		t_p approximately 5 ms (vary to obtain I_C), R_{BB1} = 120Ω, V_{BB1} = 12 V dc, R_{BB2} = ∞, V_{BB2} = 0, V_{CC} = 50 V dc, I_C = 850 mA dc, L = 100 mH 3/				
Safe operating area (switching)	3053	Clamped inductive load, T_A = +25°C, duty cycle ≤ 5 percent, t_p = 1.5 ms (vary to obtain I_C), V_{CC} = 50 V dc, I_C = 50 A dc (see figure 5)				
2N6274		Clamp voltage = 100 V dc				
2N6277		Clamp voltage = 150 V dc				
Electrical measurements		See table II, steps 1 and 2				

1/ For sampling plan, see MIL-S-19500.
2/ L = 100 µH (4 each Miller type 7827 in parallel, 40A), 0.04 ohms, or equivalent.
3/ L = 400 mH = 80 + 20 mH (2 each Triad transformer C-48u, in series), 0.713 ohms or equivalent.

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TABLE II. Groups A, B, C, and E electrical end-point measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = +1.5 V dc	I_{CEX1}			
	2N6274		V_{CE} = 120 V dc			10	µA dc
	2N6277		V_{CE} = 180 V dc			10	µA dc
2.	Forward current transfer ratio	3076	V_{CE} = 4 V dc, I_C = 20 A dc, pulsed (see 4.5.1)	h_{FE2}	30	120	
3.	Collector to emitter voltage (saturated)	3071	I_B = 20 A dc, I_C = 2 A dc, pulsed (see 4.5.1)	$\Delta V_{CE(sat)1}$	±50 mV change from initial recorded value		
4.	Thermal response 3/	3131	(See 4.3.2)	ΔV_{BE}			

1/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 2.

b. Subgroup 3, see table II herein, steps 1, 2, and 4.

c. Subgroup 6, see table II herein, steps 1, 2, and 3.

2/ The electrical measurements for table V of MIL-S-19500 are as follows:

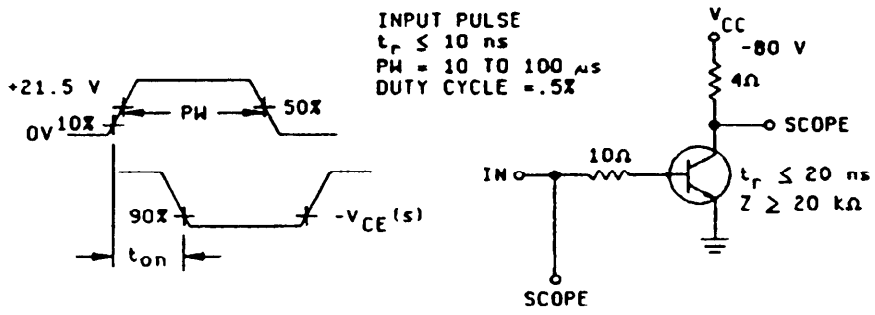
a. Subgroup 2, see table II herein, steps 1 and 2.

b. Subgroup 3, see table II herein, steps 1 and 2.

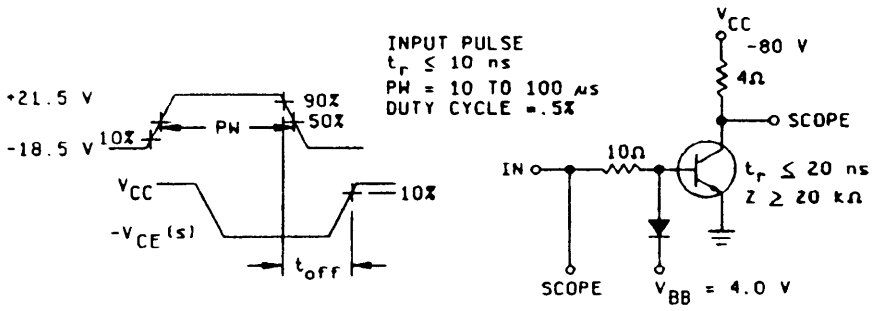
c. Subgroup 6, see table II herein, steps 1, 2, and 3.

3/ Safe operating area (SOA) testing may be performed in lieu of thermal response testing herein provided that the appropriate conditions of temperature, time, current, and voltage to achieve die attach integrity are submitted to the qualifying activity.

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Turn-on (t_{on}) time test circuit



Turn-off (t_{off}) time test circuit

FIGURE 2. Switching time test circuits.

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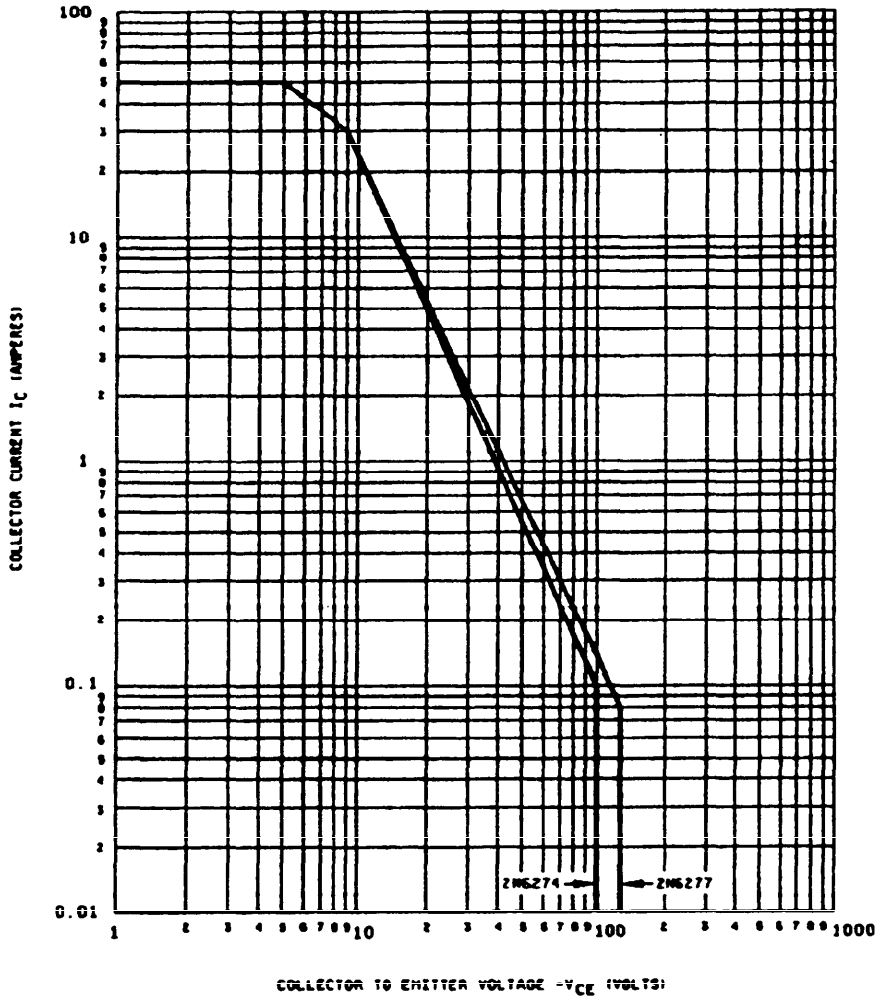


FIGURE 3. Maximum safe operating graph (continuous dc) for types 2N6274 and 2N6277.

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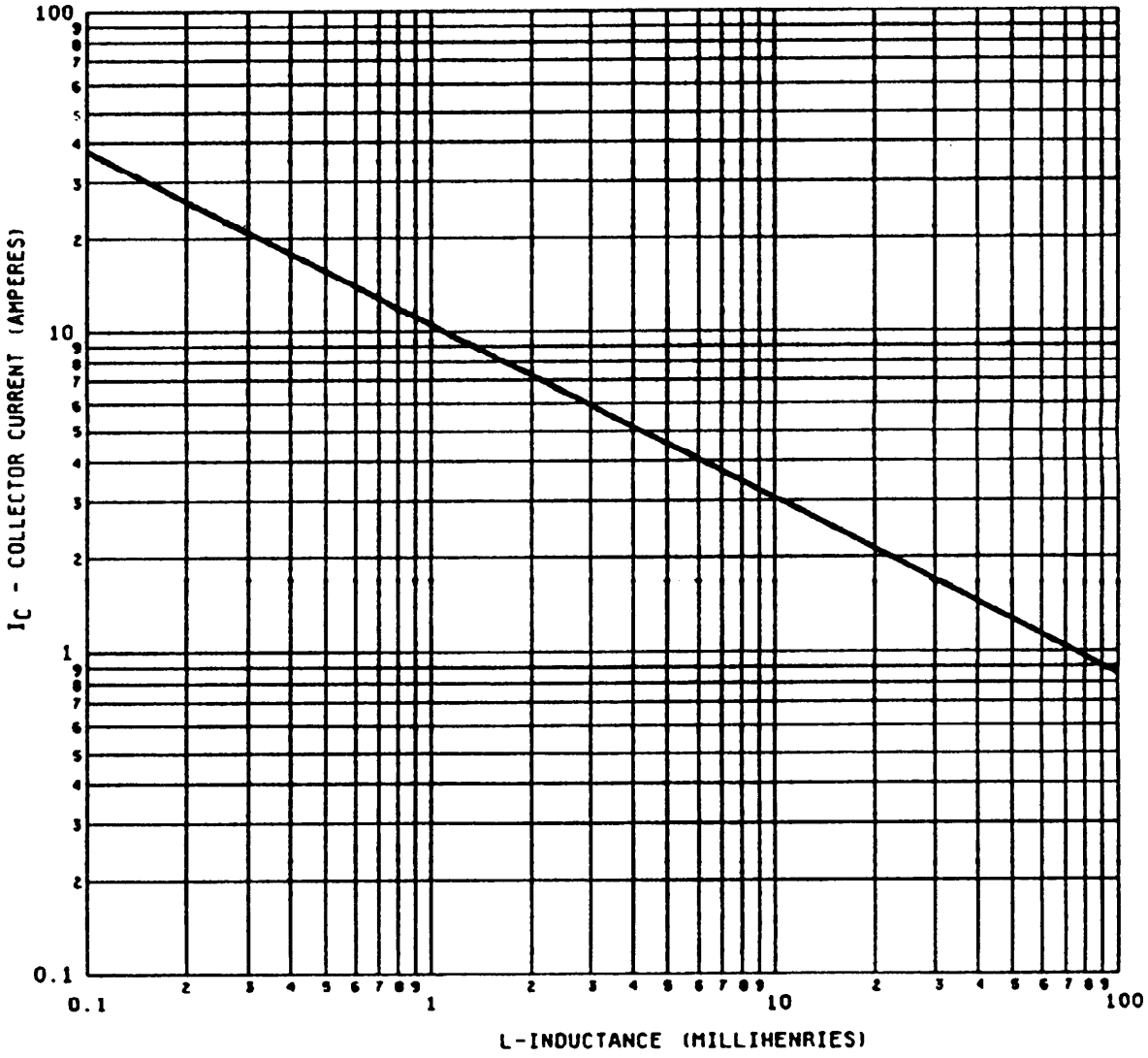
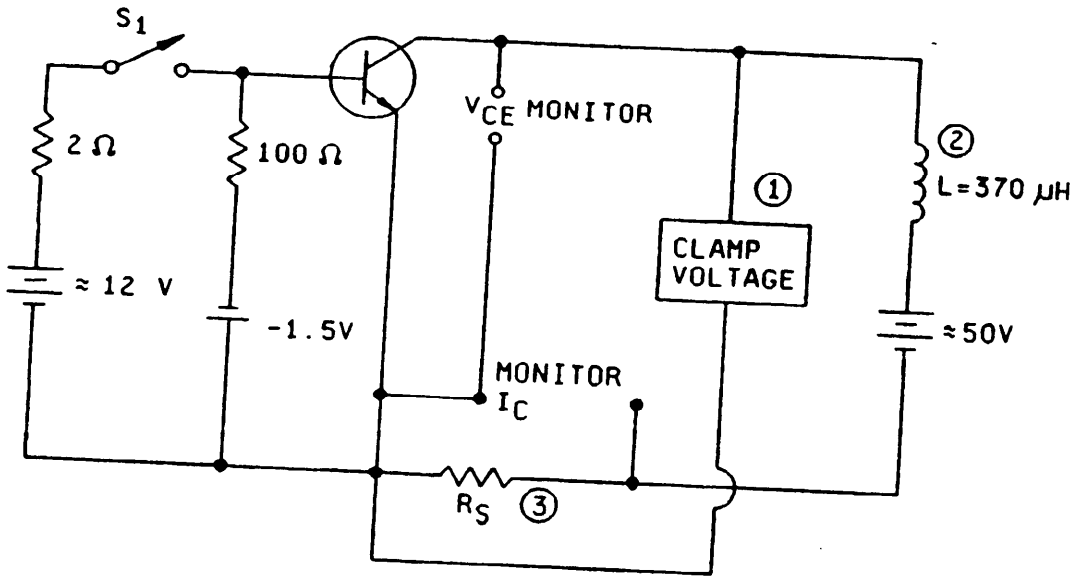


FIGURE 4. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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NOTES:
1. Either a clamping circuit or clamping diode may be used.
2. The coil used shall provide a minimum inductance of 370 μH at 50A with a maximum dc resistance of .15 ohm. For reference only: Miller type 7827; or equivalent (see 4.4.4).
3. $R_S \leq .1$ ohm, 12 W, 1 percent tolerance maximum, (noninductive).

Procedure:

1. With switch S_1 closed, set the specified test conditions.
2. Open S_1. Device fails if clamp voltage not reached and maintained until the current returns to zero.
3. Perform specified endpoint tests.

FIGURE 5. Clamped inductive sweep test circuit.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Type designation and quality assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodian:
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Navy - EC
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Review activity:
Air Force - 85

Preparing activity:
Army - ER

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	MIL-S-19500/514A	930331

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER, TYPES 2N6274 AND 2N6277 JANTX AND JANTXV

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

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